

RingNet: An Interactive Platform for Multi-Modal Data Visualization in Complex Networks

Liang Zhang (liang.zhang@tuni.fi), Xin Lai (xin.lai@tuni.fi)

Systems and Network Medicine Lab, Biomedicine Unit, Faculty of Medicine and Health Technology, Tampere University, Tampere, Finland

Introduction

Understanding cancer heterogeneity requires integrating multi-omics layers, such as gene expression, DNA methylation, single-nucleotide variants (SNV), and copy-number variations (CNV), to reveal how molecular alterations reshape cellular networks. We establish a complete workflow from multi-omics preprocessing to interactive visualization. Based on community detection in gene networks, each community represents a group of functionally related genes. Within every community, individual patients' omics data are visualized through multi-ring glyphs, where concentric rings encode different omics modalities and the outer ring represents cancer stage information. The resulting visualization allows users to intuitively explore how molecular profiles vary across patients within the same group, highlighting differences in regulatory patterns and potential disease subtypes. By integrating R-based network construction with a modern web visualization stack combining D3.js for statistical transformation, color scaling, and layout computation, and AntV G6 for high-performance interactive graph rendering, RingNet (<https://ringnet.rd.tuni.fi>) bridges computational multi-omics analysis with human interpretability. This framework provides an intuitive and scalable lens for exploring cancer heterogeneity, community-level molecular patterns, and biologically meaningful structures in disease-associated gene networks (**Fig. 1**).

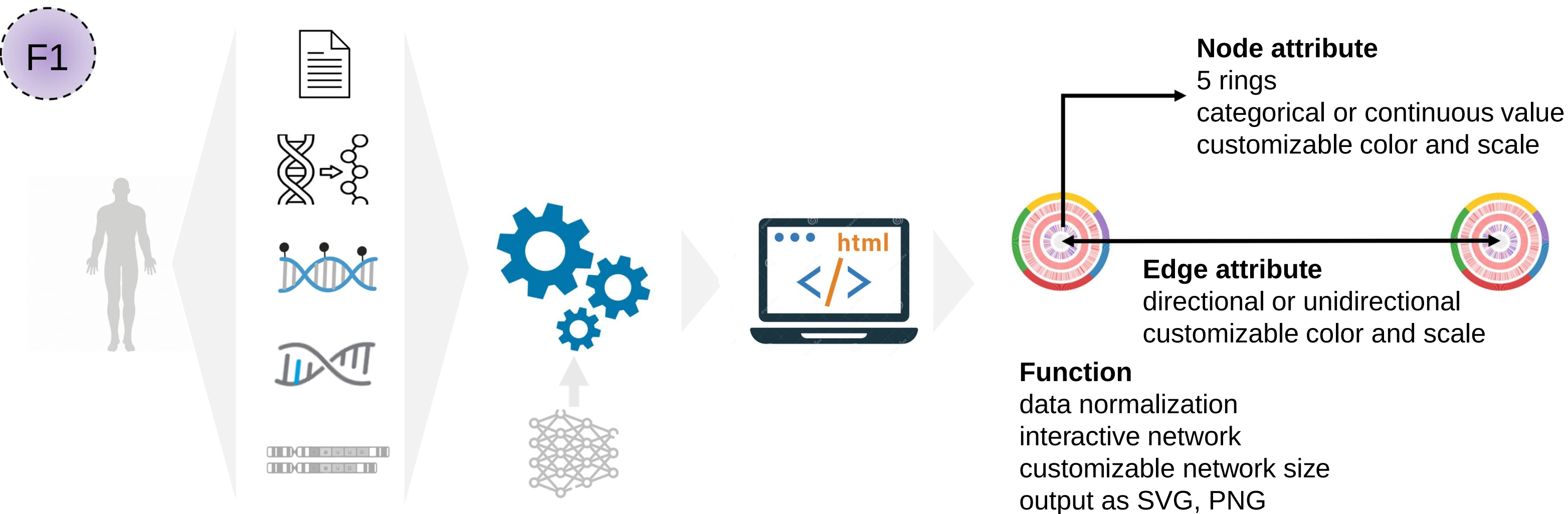


Fig. 1 Workflow of the multi-omics community-based visualization platform. Multi-omics data (expression, methylation, SNV, CNV) from cancer patients are processed in R to build gene–gene networks and detect functional communities. Each community is exported as a JSON file containing node-level omics profiles. The web module (HTML + D3.js + AntV G6) renders each gene as a multi-ring glyph, where concentric rings encode different omics layers. Interactive zooming, filtering, and dragging enable exploration of molecular patterns across patients and communities.

Methods

To enable integrative analysis and visualization of cancer multi-omics data, we developed a community-based visualization pipeline (**Fig. 2**). First, multi-omics data, including gene expression, DNA methylation, single-nucleotide variants (SNV), and copy-number variations (CNV), were processed in R to construct gene–gene interaction networks. Community detection algorithms were then applied to identify gene sets representing functionally related molecular modules. Second, after data processing, structured JSON files were generated to store node and edge information, community labels, edge weights, patient-level omics matrices, and cancer stage annotations. Third, the web-based visualization interface dynamically loads these JSON files and renders each community using an integrated D3.js and AntV G6 framework. In this framework, D3.js supports data transformation, statistical filtering, color-scale mapping, sample selection, and force-directed layout calculation, while AntV G6 provides high-performance graph rendering, node and edge visualization, dragging, zooming, focusing, and image export. Each gene is displayed as a multi-ring glyph, where concentric rings encode different omics layers, including gene expression, DNA methylation, CNV, SNV, and cancer stage. Finally, the interactive browser-based visualization enables users to intuitively explore molecular patterns, community connectivity, patient-level heterogeneity, and potential disease-associated regulatory structures.

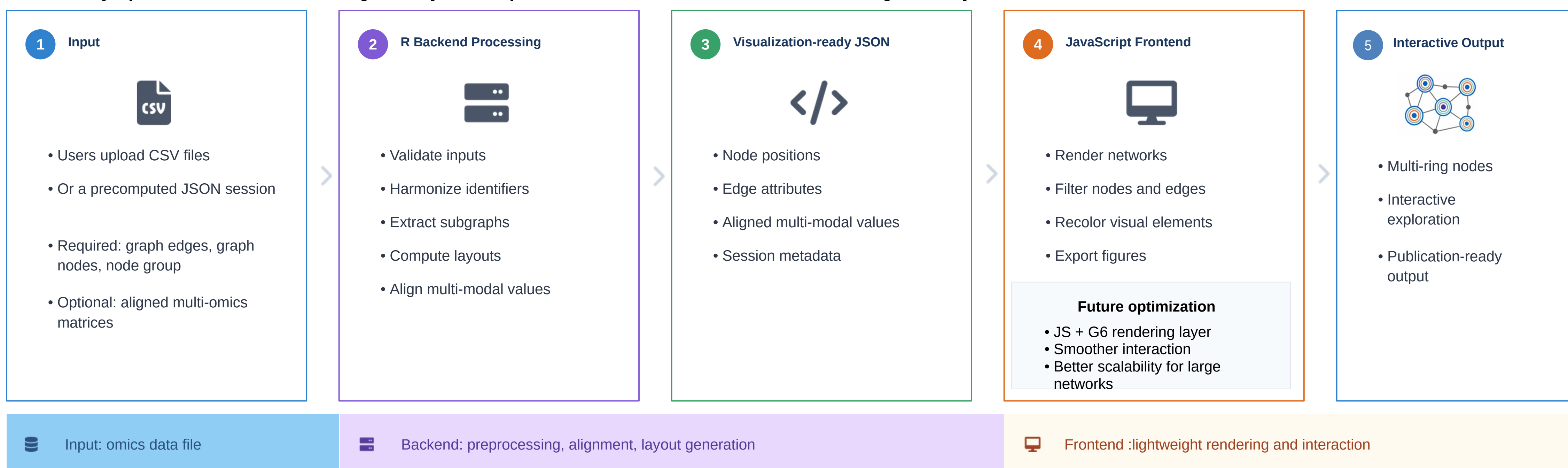


Fig. 2 Overview of the RingNet workflow. Input graph and multi-omics data are processed in R to extract community-level subnetworks and generate visualization-ready JSON files containing node attributes, edge weights, omics values, and metadata. The browser interface combines D3.js for data transformation, filtering, color mapping, sample selection, and layout calculation with AntV G6 for interactive graph rendering, zooming, dragging, focusing, and export. Genes are shown as multi-ring glyphs encoding expression, methylation, CNV, SNV, and clinical stage, while weighted edges indicate the strength and direction of gene–gene associations. Difference displaying highlights representative samples, and node-click focusing supports local subnetwork exploration.

A Case Study

OmniPath gene–gene interaction edges were integrated with TCGA breast cancer (BRCA) multi-omics datasets, including gene expression (BRCA_exp_intgr19.csv), DNA methylation (BRCA_mty_intgr19.csv), single-nucleotide variants (BRCA_snv_m.csv), and copy-number variation (BRCA_cnv_m.csv). Patient clinical stage information was obtained from BRCA_sample_stage_output.csv, and community membership was defined using BRCA_meglist_comm1.csv, which specifies the gene sets assigned to each community. In R, common genes shared across the omics layers and the interaction network were identified, and the network was partitioned into functionally related communities according to predefined community assignments. For each community, the corresponding gene–gene edges, node attributes, edge weights, patient-level omics matrices, and clinical stage annotations were extracted and exported as structured JSON files. The browser-based visualization interface then dynamically loads these JSON files and renders each community using an integrated D3.js and AntV G6 framework. D3.js is used for data transformation, statistical filtering, color-scale mapping, sample selection, and force-directed layout calculation, whereas AntV G6 provides high-performance interactive graph rendering, including node and edge visualization, dragging, zooming, focusing, and image export. Each gene is represented as a multi-ring node glyph, where concentric rings encode gene expression, DNA methylation, CNV, SNV, and clinical stage information across patients. The resulting interactive HTML visualization enables users to explore community connectivity, molecular heterogeneity, and patient-stage-associated multi-omics patterns directly within the browser.

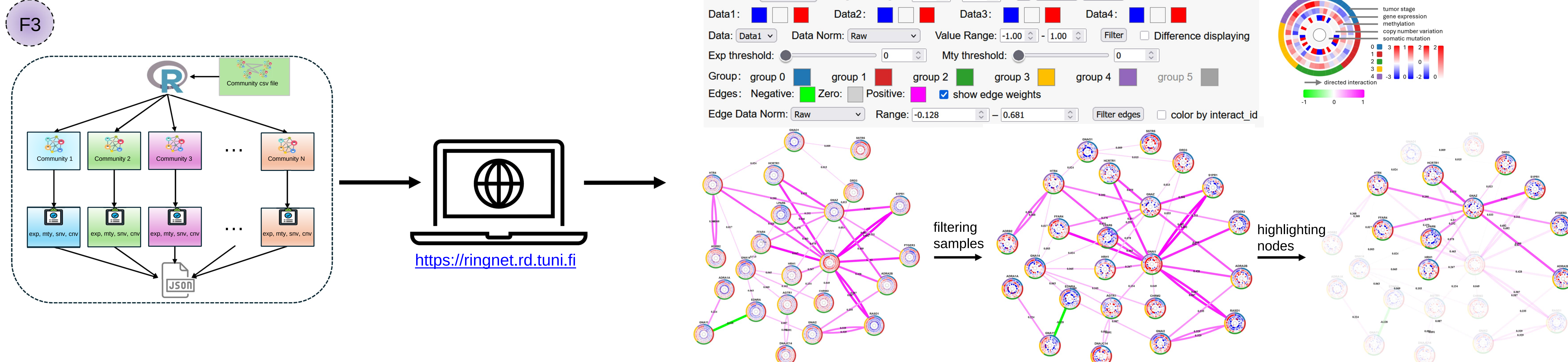


Fig. 3 Example workflow for BRCA multi-omics visualization in RingNet. Input files include community gene sets, clinical stage annotations, and multi-omics layers, including gene expression, DNA methylation, SNV, and CNV. The RingNet web interface allows users to customize node colors, displayed value ranges, and edge attributes, and to export results as SVG, PNG files. The R module extracts community-level omics matrices, node information, gene–gene interactions, edge weights, and clinical annotations into structured JSON files. In the browser, each gene is shown as a multi-ring node glyph encoding different omics layers across patients, while edges represent gene–gene interactions weighted by correlation coefficients. The Difference displaying function selects a shared subset of representative samples to highlight the most informative multi-omics trends across nodes, enabling intuitive exploration of weighted network structure and patient-level molecular heterogeneity.

Summary

In summary, RingNet provides a community-based visualization framework for integrating and interpreting cancer multi-omics networks. By combining OmniPath interaction networks, TCGA/GTEX omics data, R-based community detection, and a browser-based D3.js + AntV G6 visualization interface, RingNet enables interactive exploration of patient-level molecular heterogeneity within functionally related gene modules. Each gene is represented as a multi-ring glyph encoding expression, DNA methylation, CNV, SNV, and clinical stage, while weighted edges indicate the strength and direction of gene–gene associations. The Difference displaying function highlights representative patient samples with informative multi-omics trends, and node-click focusing allows users to inspect local subnetworks while de-emphasizing unrelated nodes. Applied to BRCA, RingNet revealed stage-associated molecular patterns and demonstrated its potential as a scalable tool for discovering biologically meaningful structures in cancer multi-omics networks.